EIN 4343 Inventory and Supply Chain Systems

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

FORECASTING: PRELIMINARIES

The role of forecasting

- Process of estimating the volume of demand or service to handled or provided by the firm

- Needed for efficient and effective plan and control of activities

Air Garnetts vs. Air Jordans

- i2 Techologies -- supply chain software provider for Nike in addition to Siemens, Kmart and Caterpillar

- A glitch in the forecasting module caused Nike to send orders for thousands more of Air Garnetts and thousands fewer of Air Jordans to the manufacturing plants in 2,000
  - $100M in lost sales!

https://www.cio.com/article/2439601/nike-rebounds--how--and-why--nike-recovered-from-its-supply-chain-disaster.html

Issues related to demand forecasting

- Spatial vs. Temporal Demand
- Regular vs. Intermittent Demand
- Independent vs. Dependent Demand

Tale of two companies

Sports Obermeyer

- Seasonal demand
- Fashion product
- Historical data may or may not be useful

Barilla Spa

- Stationary demand
- Consumer product
- Historical data is useful

Characteristics of forecasts

1. Forecasts are always inaccurate and should thus include both the expected value of the forecast and a measure of forecast error

Characteristics of forecasts

2. Long-term forecasts are usually less accurate than short-term forecasts

Characteristics of forecasts

3. Aggregate forecasts are usually more accurate than disaggregate forecasts

Characteristics of forecasts

4. In general, the farther up the supply chain a company is, the greater is the distortion of information it receives

Forecasting methods

Qualitative methods

- Primarily subjective and rely on judgment
- Different stakeholders
  - Sales force composite
  - Market research
  - Jury of executive opinion
  - Delphi Method

Forecasting methods

Quantitative methods

- Rely on data
- Different approaches
  1. Time Series Methods
     - Historical demand data
  2. Causal Methods
     - Relationship between demand and some other factor
  3. Simulation Approaches
     - Imitate consumer choices that give rise to demand

DATA ANALYTICS

Demand forecasting

Observed demand = systematic component + random component

1. Systematic component – underlying behavior that can be detected and exploited to inform decision making

- ***Stationary*** (base demand)
- ***Trend*** (growth or decline in demand)
- ***Cyclicality or Seasonality*** (predictable patterns of fluctuation)

2. Random component – noise that will always be there

Notation

T: planning horizon indexed by $t = 1, \ldots, T$

D_t: observed demand in period t

F_t: forecasted demand for period t

Forecast error: What?

- Each demand observation has the systematic component and a random component.
- A good forecasting method captures the systematic component.
- The random component can be attributed to forecast error.

Issues related to demand forecasting

- Spatial vs. Temporal Demand
- Regular vs. Intermittent Demand
- Independent vs. Dependent Demand

Forecast error: Why?

- Serves two purposes
  1. Whether the current forecasting method predicts the systematic component accurately.
  2. All contingency plans for the supply chain must account for forecast error.

- For a time period t, the *forecast error* is given by

$$E_t = F_t - D_t$$

Measures of forecast error

- Mean squared error $$MSE_n = \frac{1}{n}\sum_{t=1}^{n} E_t^2$$
- Mean absolute deviation $$MAD_n = \frac{1}{n}\sum_{t=1}^{n} |E_t|$$
- Mean absolute percentage error $$MAPE_n = \frac{1}{n}\sum_{t=1}^{n} \left|\frac{E_t}{D_t}\right| 100$$

Measures of forecast error: MSE

Mean squared error $MSE_n = \frac{1}{n}\sum_{t=1}^{n} E_t^2$

- MSE relates to the variance of forecast error. Typically, we assume that the random component of demand has a mean of 0 and variance of MSE.

- MSE penalizes large errors much more significantly than small errors because all errors are squared.

- Use MSE if the cost of a large error is much larger than the gains from very accurate forecasts.

Measures of forecast error: MAD

Mean absolute deviation $MAD_n = \frac{1}{n}\sum_{t=1}^{n} |E_t|$

- MAD is appropriate when the cost of forecasting error is proportional to the size of the error.

- MAD is a better measure than MSE if the error distribution is non-symmetric.

- MAD can be used to estimate the variance of the random component using $\sigma = 1.25MAD$ assuming that the random component is normally distributed.

Measures of forecast error: MAPE

Mean absolute percentage error $MAPE_n = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{E_t}{D_t}\right| 100$

- MAPE is a good measure when the underlying forecast has seasonality and demand varies considerably between periods.

Forecasting: Methods

- Stationary series
  - Moving Averages
  - Single Exponential Smoothing

$$D_t = \boxed{\mu} + \epsilon_t$$

- Linear trend series
  - Holt's Double Exponential Smoothing
  - Linear Regression Analysis

$$D_t = \boxed{\mu} + \boxed{\lambda} t + \epsilon_t$$

- Seasonal series
  - Winter's Triple Exponential Smoothing

$$D_t = \boxed{\gamma}(\boxed{\mu} + \boxed{\lambda} t) + \epsilon_t$$

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FORECASTING: STATIONARY SERIES

Stationary series

If we are not expecting the demand to change substantially, what is a good approach to obtain a demand forecast?

How much of the historical data should be used?

Stationary series

An observation from a stationary time series can be expressed as follows

$$D_t = \boxed{\mu} + \boxed{\epsilon_t}$$

Random error term with a mean value of 0 and constant variance σ^2

Unknown population mean of the underlying process

Moving Averages (MA)

Given N past demand observations $D_{t-N}, D_{t-N+1}, \ldots D_{t-1}$, the demand forecast F_t for period t obtained at the end of period $t-1$ given by,

$$F_t = \frac{1}{\boxed{N}} \sum_{i=t-N}^{t-1} D_i$$

Sample size

Example 9.1: MA

Weekly data for pounds of coffee used at a local coffee shop during the last two months are 27, 21, 27, 30, 24, 24, 27, and 21.

If (a) three-week (b) four-week MA forecasts are used for forecasting, determine <u>one-step-ahead</u> forecasts.

How can we obtain <u>multiple step-ahead</u> forecasts? Suppose that we are at the end of period 3 (that is we have already observed the demand for period 3) and we want to obtain a MA(3) forecast for period 6.

Example 9.1: MA

$D = (27,21,27,30,24,24,27,21)$

a. Three-week MA forecasts

$$F_{3,4} = \frac{27 + 21 + 27}{3} = 25 \text{ units}$$

$$F_{4,5} = \frac{21 + 27 + 30}{3} = 26 \text{ units}$$

$$F_{5,6} = \frac{27 + 30 + 24}{3} = 27 \text{ units}$$

$$F_{6,7} = \frac{30 + 24 + 24}{3} = 26 \text{ units}$$

$$F_{7,8} = \frac{24 + 24 + 27}{3} = 25 \text{ units}$$

$$F_{8,9} = \frac{24 + 27 + 21}{3} = 24 \text{ units} = F_{8,10} = F_{8,11} = F_{8,12} = \cdots = F_{8,T}$$

Single-step ahead forecasts

Multiple-step ahead forecasts

Example 9.1: MA

$D = (27,21,27,30,24,24,27,21)$

b. Four-week MA forecasts

$$F_{4,5} = \frac{27 + 21 + 27 + 30}{4} = 26.25 \text{ units}$$

$$F_{5,6} = \frac{21 + 27 + 30 + 24}{4} = 25.5 \text{ units}$$

$$F_{6,7} = \frac{27 + 30 + 24 + 24}{4} = 26.25 \text{ units}$$

$$F_{7,8} = \frac{30 + 24 + 24 + 27}{4} = 26.25 \text{ units}$$

$$F_{8,9} = \frac{24 + 24 + 27 + 21}{4} = 24 \text{ units} = F_{8,10} = F_{8,11} = F_{8,12} = \cdots = F_{8,T}$$

Single-step ahead forecasts

Multiple-step ahead forecasts

Example 9.1: MA: Which one?

Time (t)	Demand (D_t)	Forecast ($F_t^{MA(3)}$)	Error (E_t)	Absolute Error ($\lvert E_t \rvert$)
1	27			
2	21			
3	27			
4	30	25	-5	5
5	24	26	2	2
6	24	27	3	3
7	27	26	-1	1
8	21	25	4	4
9		24		

$$MAD(N=3) = \frac{5+2+3+1+4}{5} = \frac{15}{5} = 3$$

Example 9.1: MA: Which one?

Time (t)	Demand (D_t)	Forecast ($F_t^{MA(4)}$)	Error (E_t)	Absolute Error ($\lvert E_t \rvert$)
1	27			
2	21			
3	27			
4	30			
5	24	26.25	2.25	2.25
6	24	25.5	1.5	1.5
7	27	26.25	-0.75	0.75
8	21	26.25	5.25	5.25
9		24		

$$MAD(N=4) = \frac{2.25 + 1.5 + 0.75 + 5.25}{4} = \frac{9.75}{4} = 2.44$$

Example 9.1: MA: Which one?

$$MAD(N=3) = \frac{5+2+3+1+4}{5} = \frac{15}{5} = 3$$

$$MAD(N=4) = \frac{2.25+1.5+0.75+5.25}{4} = \frac{9.75}{4} = 2.44$$

Using MAD, MA(4) appears to be a slightly better approach. However, given the size of the sample, the difference might not be statistically significant.

Simple Exponential Smoothing (SES)

Given the most recent demand observation D_{t-1}; the most recent demand forecast F_{t-1} and the so-called smoothing constant $0 < \alpha < 1$, demand forecast F_t obtained at the end of period $t - 1$ is given by,

$$F_t = \boxed{\alpha} D_{t-1} + (1 - \alpha) F_{t-1}$$

Relative weight given to the most recent demand observation, i.e., the smoothing constant

Simple Exponential Smoothing (SES)

Another way to look at it….

$$F_t = \alpha D_{t-1} + (1-\alpha)F_{t-1}$$

$$= \alpha D_{t-1} + F_{t-1} - \alpha F_{t-1}$$

$$= F_{t-1} + \boxed{\alpha(D_{t-1} - F_{t-1})}$$

Forecast error in the previous period!

Under SES, next forecast is the previous forecast adjusted by the error associated with the previous forecast

Simple Exponential Smoothing (SES)

Yet, another way to look at it….

$$F_t = \alpha D_{t-1} + (1-\alpha)F_{t-1}$$

$$F_{t-1} = \boxed{\alpha D_{t-2} + (1-\alpha)F_{t-2}}$$

$$F_{t-2} = \boxed{\alpha D_{t-3} + (1-\alpha)F_{t-3}}$$

$$F_t = \sum_{i=0}^{\infty} \alpha(1-\alpha)^i D_{t-i-1}$$

Under SES, a forecast is a weighted average of all previous demand observations!

Example 9.2: SES

Weekly data for pounds of coffee used at a local coffee shop during the last two months are 27, 21, 27, 30, 24, 24, 27, and 21.

Assume that the forecast made for the first week at the end of week zero was 22 pounds.

Determine one-step ahead forecasts for weeks 2 through 9 using single exponential smoothing with (a) a smoothing constant of 0.1 and (b) with a smoothing constant of 0.25. Which of the forecasting approaches should be used? Please justify your answer.

Example 9.2: SES

$D = (27,21,27,30,24,24,27,21)$

a. SES forecasts using a smoothing factor of 0.10

$F_{1,2} = (0.10)(27) + (1 - 0.10)(22) = 22.50 \text{ units}$

$F_{2,3} = (0.10)(21) + (1 - 0.10)(22.5) = 22.35 \text{ units}$

$F_{3,4} = (0.10)(27) + (1 - 0.10)(22.35) = 22.40 \text{ units}$

$F_{4,5} = (0.10)(30) + (1 - 0.10)(22.82) = 23.53 \text{ units}$

$F_{5,6} = (0.10)(24) + (1 - 0.10)(23.53) = 23.58 \text{ units}$

$F_{6,7} = (0.10)(24) + (1 - 0.10)(23.58) = 23.62 \text{units}$

$F_{7,8} = (0.10)(27) + (1 - 0.10)(23.62) = 23.96 \text{ units}$

$F_{8,9} = (0.10)(21) + (1 - 0.10)(23.96) = 23.66 \text{ units} = F_{8,10} = F_{8,11} = \cdots = F_{8,T}$

Multiple-step ahead forecasts

Single-step ahead forecasts

Example 9.2: SES

$D = (27,21,27,30,24,24,27,21)$

a. SES forecasts using a smoothing factor of 0.25

$F_{1,2} = (0.25)(27) + (1 - 0.25)(22) = 23.25 \text{ units}$

$F_{2,3} = (0.25)(21) + (1 - 0.25)(23.25) = 22.69 \text{ units}$

$F_{3,4} = (0.25)(27) + (1 - 0.25)(22.69) = 23.77 \text{ units}$

$F_{4,5} = (0.25)(30) + (1 - 0.25)(23.77) = 25.32 \text{ units}$

$F_{5,6} = (0.25)(24) + (1 - 0.25)(25.32) = 24.99 \text{ units}$

$F_{6,7} = (0.25)(24) + (1 - 0.25)(24.99) = 24.74 \text{ units}$

$F_{7,8} = (0.25)(27) + (1 - 0.25)(24.74) = 25.31 \text{ units}$

$F_{8,9} = (0.25)(21) + (1 - 0.25)(25.31) = 24.23 \text{ units} = F_{8,10} = F_{8,11} = \cdots = F_{8,T}$

Multiple-step ahead forecasts

Single-step ahead forecasts

Example 9.2: SES: Which one?

Time (t)	Demand (D_t)	Forecast ($F_t^{SES(0.1)}$)	Error (using MSE)
1	27	22	$(27-22)^2$
2	21	22.50	$(21-22.50)^2$
3	27	22.35	$(27-22.35)^2$
4	30	22.82	$(30-22.82)^2$
5	24	23.53	$(24-23.53)^2$
6	24	23.58	$(24-23.58)^2$
7	27	23.62	$(27-23.62)^2$
8	21	23.96	$(21-23.96)^2$
9		23.66	

$$MSE(\alpha = 0.1) = \frac{121.06}{8} = 15.13$$

Example 9.2: SES: Which one?

Time (t)	Demand (D_t)	Forecast ($F_t^{SES(0.25)}$)	Error (using MSE)
1	27	22	$(27-22)^2$
2	21	23.25	$(21-23.25)^2$
3	27	22.69	$(27-22.69)^2$
4	30	23.77	$(30-23.77)^2$
5	24	25.32	$(24-25.32)^2$
6	24	24.99	$(24-24.99)^2$
7	27	24.74	$(27-24.74)^2$
8	21	25.31	$(21-25.31)^2$
9		24.23	

$$MSE(\alpha = 0.25) = \frac{113.92}{8} = 14.24$$

Example 9.2: SES: Which one?

$$MSE(\alpha = 0.1) = \frac{121.06}{8} = 15.13$$

$$MSE(\alpha = 0.25) = \frac{113.92}{8} = 14.24$$

Using MSE, SES(0.25) appears to be a slightly better approach. However, given the size of the sample, the difference might not statistically significant.

Forecasting: Methods

- Stationary series
  - Moving Averages
  - Single Exponential Smoothing

$$D_t = \mu + \epsilon_t$$

- Linear trend series
  - Holt's Double Exponential Smoothing
  - Linear Regression Analysis

$$D_t = \mu + \lambda t + \epsilon_t$$

- Seasonal series
  - Winter's Triple Exponential Smoothing

$$D_t = \gamma(\mu + \lambda t) + \epsilon_t$$

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FORECASTING: LINEAR TREND SERIES

Linear trend series

If we are expecting the demand to change linearly, what is a good approach to obtain a demand forecast?

How much of the historical data should be used?

Double exponential smoothing: Holt's

Let us define some new parameters

S_t	value of the intercept at the end of period t
G_t	value of the slope at the end of period t
α, β	smoothing constants

in addition to the ones from before

D_t	demand observed at the end of period t
$F_{t-1,t}$	demand forecast for period t at the end of period $t-1$

Holt's

Given the most recent demand observation and the intercept and slope for the most recent demand forecast, the forecast parameters are updated using

$$S_t = \alpha D_t + (1 - \alpha)(S_{t-1} + G_{t-1})$$

$$G_t = \beta(S_t - S_{t-1}) + (1 - \beta)G_{t-1}$$

Smoothing constants $0 < \alpha, \beta < 1$

Holt's

Consequently, the demand forecast obtained at the end of period t given by,

$$F_{t,t+1} = S_t + 1G_t$$

for one-step ahead forecast

$$F_{t,t+\pi} = S_t + \pi G_t$$

and for multiple-step ahead forecast where π is the number of periods in the future.

Example 9.3: Holt's

Banana Cafe in Key West, Florida is a premier bakery that sells pastries. Suppose that you are hired as a consultant to help the store to determine the product mix for the coming months. The number of strawberry croissants sold per month are given as follows: 170, 176, 204 and 240 units. Suppose that the initial estimates for the intercept and the slope are 110 and 20 units, respectively.

Provide the demand forecasts for the forecasts for months 5 through 8 for strawberry croissants using $\alpha = 0.25$ and $\beta = 0.15$

Example 9.3 Holt's DES

$D = (170,176,204,240)$

DES forecasts using a smoothing factors of 0.25 and 0.15

$$S_1 = (0.25)(170) + (1 - 0.25)(110 + 20) = \boxed{140 \text{ units}}$$

$$G_1 = (0.15)(140 - 110) + (1 - 0.15)(20) = \boxed{21.5 \text{ units}}$$

$$S_2 = (0.25)(176) + (1 - 0.25)(140 + 21.5) = \boxed{165.13 \text{ units}}$$

$$G_2 = (0.15)(165.13 - 140) + (1 - 0.15)(21.5) = \boxed{22.04 \text{ units}}$$

$$S_3 = (0.25)(204) + (1 - 0.25)(165.13 + 22.04) = \boxed{191.38 \text{ units}}$$

$$G_3 = (0.15)(191.38 - 165.13) + (1 - 0.15)(22.04) = \boxed{22.67 \text{ units}}$$

$$S_4 = (0.25)(240) + (1 - 0.25)(191.38 + 22.67) = \boxed{220.54 \text{ units}}$$

$$G_4 = (0.15)(220.54 - 191.38) + (1 - 0.15)(22.67) = \boxed{23.65 \text{ units}}$$

Example 9.3 Holt's DES

We have

$$S_4 = (0.25)(240) + (1 - 0.25)(191.38 + 22.67) = 220.54 \text{ units}$$

$$G_4 = (0.15)(220.54 - 191.38) + (1 - 0.15)(22.67) = 23.65 \text{ units}$$

Consequently, we have

$$F_{4,4+1} = 220.54 + (1)(23.65) = 244.19 \text{ units}$$

$$F_{4,4+2} = 220.54 + (2)(23.65) = 267.84 \text{ units}$$

$$F_{4,4+3} = 220.54 + (3)(23.65) = 291.49 \text{ units}$$

$$F_{4,4+4} = 220.54 + (4)(23.65) = 315.45 \text{ units}$$

Linear regression analysis (LRA)

The objective of LRA is to determine the straight line that best represents the data set. That is, the deviation between the actual demand observation and the corresponding demand forecast, i.e., $(D_t - F_t)$, should be as small as possible for the forecasts to be accurate.

Suppose we have a sample T pairs of demand observations and the corresponding forecasts. We can construct an optimization model to minimize the total the squared deviation between the demand and the forecasts, which is given by

$$\sum_{t=1}^{T} (D_t - F_t)^2$$

Linear regression analysis (LRA)

The model can be rewritten as

$$Z(\hat{a}, \hat{b}) = \sum_{t=1}^{T} \left(D_t - (\hat{a} + \hat{b}t)\right)^2$$

Then, the problem can be stated as

$$\min \; Z(\hat{a}, \hat{b})$$

Note that this is an unconstrained continuous optimization model with two decision variables.

Linear regression analysis (LRA)

First order conditions are given by

$$\frac{\partial Z}{\partial \hat{a}} = \sum_{t=1}^{T} 2(D_t - \hat{a} - \hat{b}t)(-1)$$

$$\frac{\partial Z}{\partial \hat{b}} = \sum_{t=1}^{T} 2(D_t - \hat{a} - \hat{b}t)(-t)$$

Solving for the estimators simultaneously yields

$$\hat{b} = \frac{\sum_{t=1}^{T}(t-\bar{t})(D_t - \bar{D})}{\sum_{t=1}^{T}(t-\bar{t})^2} = \frac{T\sum_{t=1}^{T} D_t t - \left(\sum_{t=1}^{T} D_t\right)\left(\sum_{t=1}^{T} t\right)}{T\sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2}$$

$$\hat{a} = \bar{D} - \hat{b}\bar{t}$$

Linear regression analysis (LRA)

where

$$\overline{D} = \frac{\sum_{t=1}^{T} D_t}{T}$$

$$\overline{T} = \frac{\sum_{t=1}^{T} t}{T}$$

$$\sum_{t=1}^{T} t = \frac{T(T+1)}{2}$$

$$\sum_{t=1}^{T} t^2 = \frac{T(T+1)(2T+1)}{6}$$

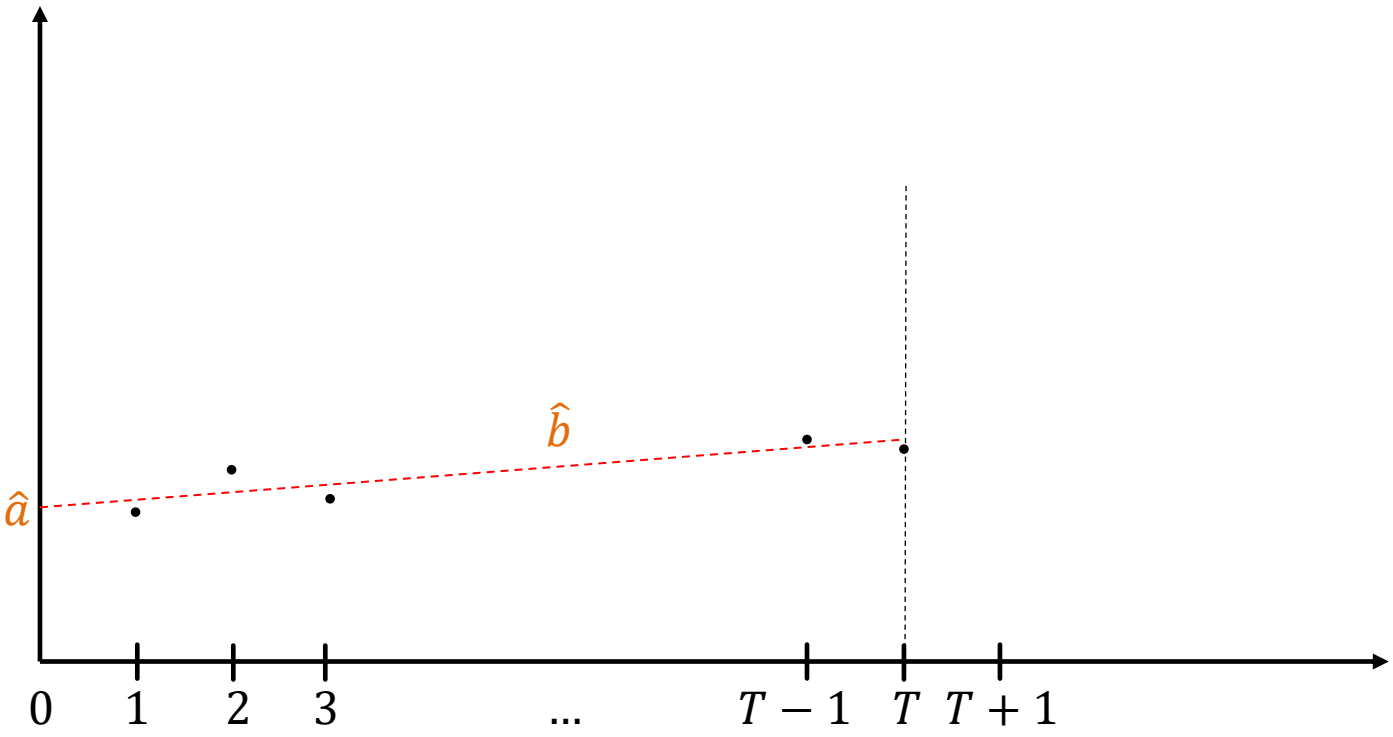
Linear regression analysis (LRA)

Consequently, the demand forecasts obtained at the end of period T are given by

$$F_{T,T+1} = \hat{a} + (T+1)\hat{b}$$

$$F_{T,T+\pi} = \hat{a} + (T+\pi)\hat{b}$$

Linear regression analysis (LRA)



Note that the output of LRA which are the intercept and slope estimates are obtained for $t = 0$.

Example 9.3: LRA

Banana Cafe in Key West, Florida is a premier bakery that sells pastries. Suppose that you are hired as a consultant to help the store to determine the product mix for the coming months. The number of strawberry croissants sold per month are given as follows: 170, 176, 204 and 240 units.

Provide the demand forecasts for the forecasts for months 5 through 8 for strawberry croissants using LRA.

Example 9.3: LRA

$$\bar{D} = \frac{\sum_{t=1}^{T} D_t}{T} = \frac{170 + 176 + 204 + 240}{4} = \frac{790}{4} = 197.5$$

$$\bar{T} = \frac{\sum_{t=1}^{T} t}{T} = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

$$\sum_{t=1}^{T} t = \frac{T(T+1)}{2} = \frac{(4)(4+1)}{2} = 10$$

$$\sum_{t=1}^{T} t^2 = \frac{T(T+1)(2T+1)}{6} = \frac{(4)(4+1)((2)(4)+1)}{6} = 30$$

$$\hat{b} = \frac{T\sum_{t=1}^{T} D_t t - \left(\sum_{t=1}^{T} D_t\right)\left(\sum_{t=1}^{T} t\right)}{T\sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2} = \frac{(4)[(1)(170) + \cdots + (4)(240)] - (790)(10)}{(4)(30) - (10)^2} = 23.8$$

$$\hat{a} = \bar{D} - \hat{b}\bar{t} = 197.5 - (23.8)(2.5) \quad = 138$$

Example 9.3: LRA

$$F_{4,4+1} = 138 + (4+1)(23.8) = 257$$

$$F_{4,4+2} = 138 + (4+2)(23.8) = 280.8$$

$$F_{4,4+3} = 138 + (4+3)(23.8) = 304.6$$

$$F_{4,4+4} = 138 + (4+4)(23.8) = 328.4$$